



BAXI 2213: Knowledge Based System

KBS Life Cycle:

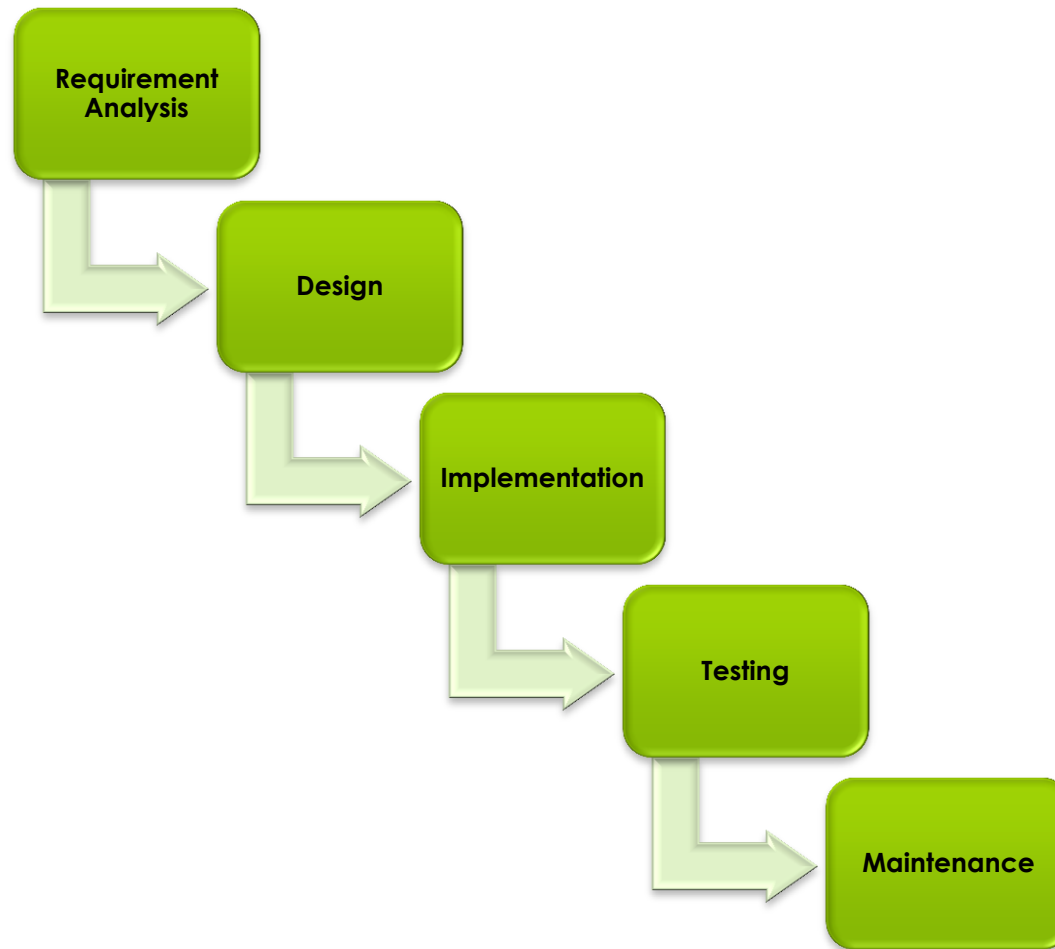
1. Waterfall model of software development
2. KBS lifecycle
3. Knowledge Acquisition

Waterfall Model of Software Development



Knowledge Based System

Waterfall Model



Knowlegde Based System

Type of Knowledge Represented

Software Engineering (SE)

- SE represents **well-known and well-defined algorithmic procedures** known by people.

Knowledge Engineering (KE)

- KE represents **extensive, imprecise, and ill-define heuristic knowledge** that is stored in experts' mind.

Knowlegde Based System

Nature and Quantity of Knowledge

Software Engineering (SE)

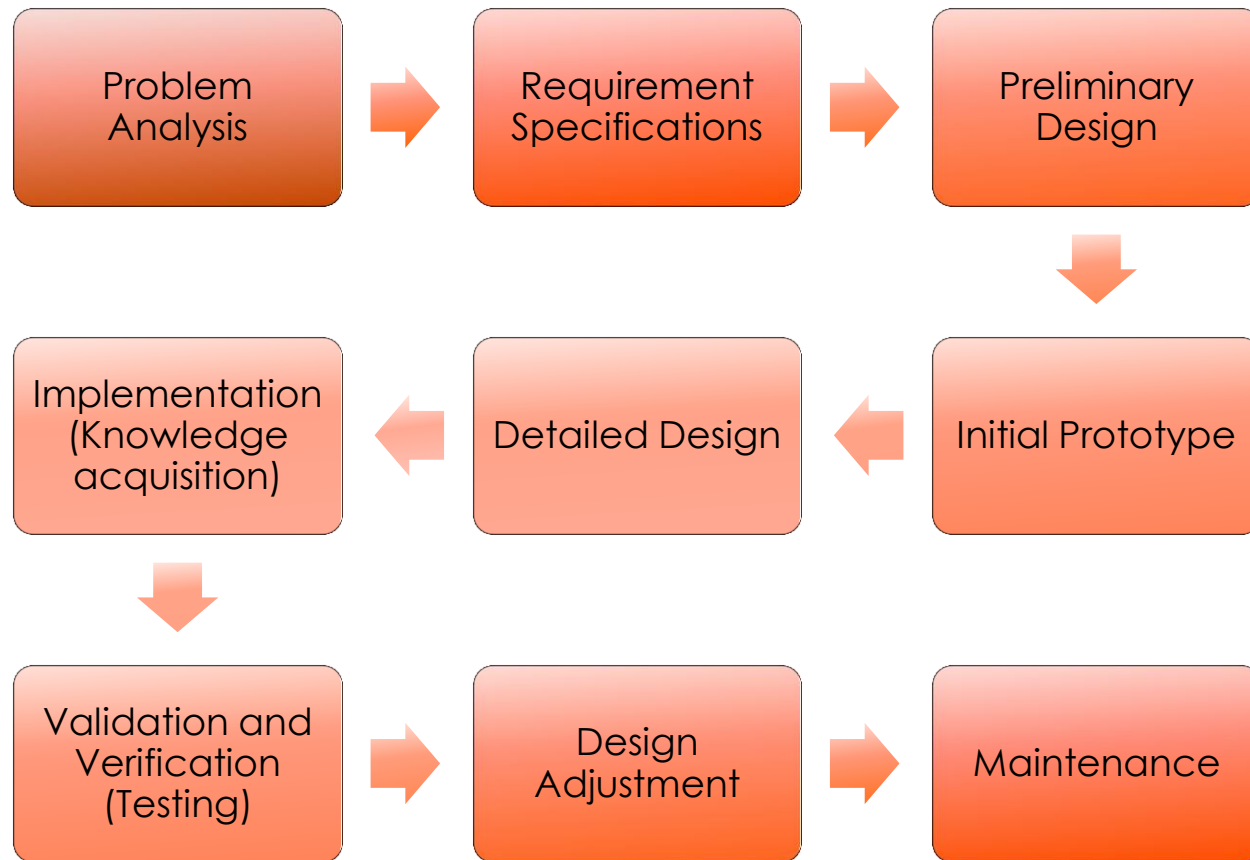
- The nature and quantity of knowledge required **can be estimated reasonably well.**

Knowledge Engineering (KE)

- The nature and quantity of knowledge **is not well known even by the experts.**

Knowledge Based System

KBS Life Cycle Model



Knowlegde Based System

KBS Life Cycle Model

Problem Analysis

- **Evaluate the problem and resources:** to determine the applicability of a knowledge-based solution.
- **Determine cost and benefits:** to ascertain whether its development is warranted.
- **Requires either market research:** to determine the cost effectiveness of the system.

KBS Life Cycle Model

Requirement Specifications

- **Formalize and write** the analysis stage.
- **Sets the objectives.**
- **Outlines the means** to obtain these objectives.
- **Discuss the objectives** and the features of the system, its user environment, and its constraints.

KBS Life Cycle Model

Preliminary Design

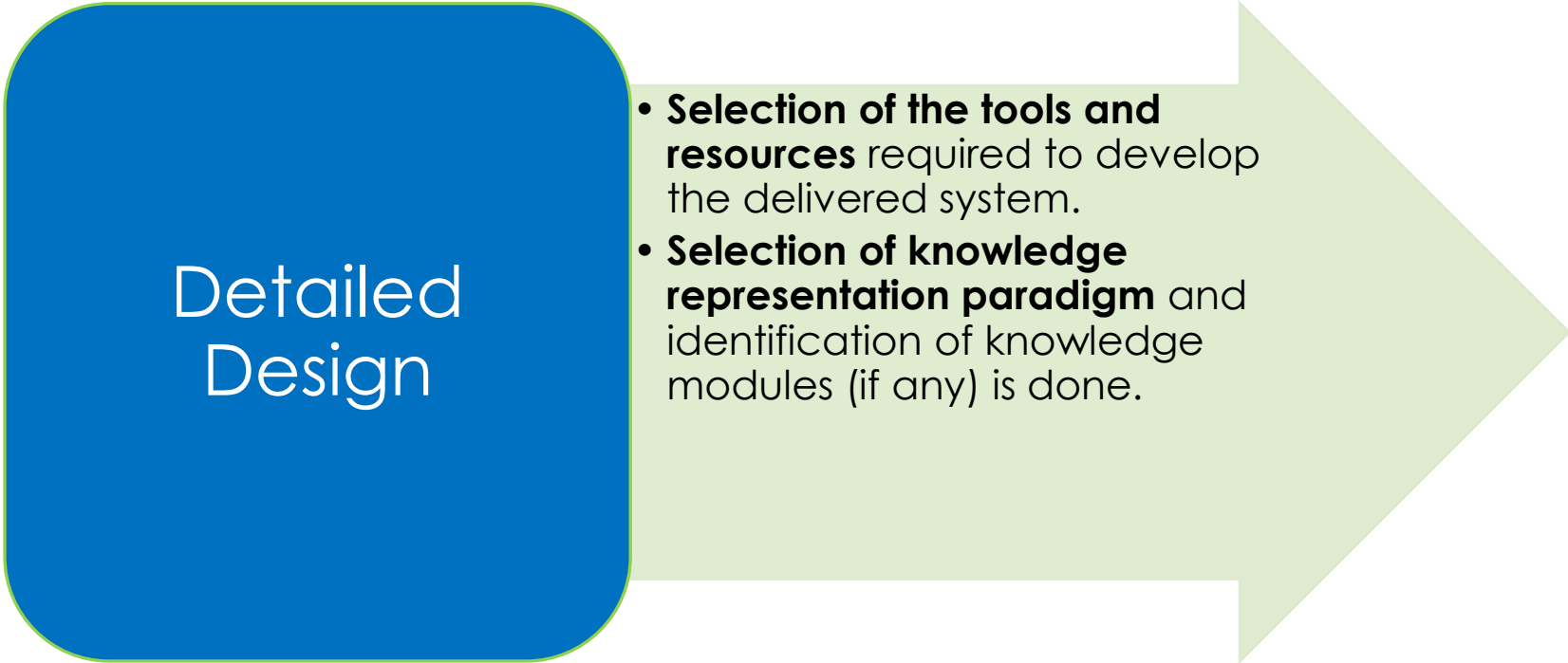
- **Deals with the high level decisions** to begin the initial prototype.
- **Determines the KR paradigm**, the tool chosen for the prototype, and the selection of experts.
- **Considerable acquisition of general knowledge** from experts or from printed sources.

KBS Life Cycle Model

Initial Prototype

- The most important stage.
- **Justified or overturned based on the knowledge elicited** from the expert.
- Should be look like **the complete system** except that it will be limited in breadth.
- Should include **a fairly well-developed user interface** and a reasonably robust knowledge subset.

KBS Life Cycle Model



Detailed Design

The diagram illustrates the 'Detailed Design' phase of the KBS Life Cycle Model. It features a large blue rounded rectangle on the left containing the text 'Detailed Design'. To its right is a large light green arrow pointing to the right, which contains a bulleted list of tasks. The entire slide is framed by a green border with a geometric pattern.

- **Selection of the tools and resources** required to develop the delivered system.
- **Selection of knowledge representation paradigm** and identification of knowledge modules (if any) is done.

KBS Life Cycle Model

Implementation
(Knowledge
Acquisition)

- The **most time consuming** of the KBS lifecycle.
- Implementation includes the **complete knowledge acquisition process for all subsystems.**

KBS Life Cycle Model

Validation and Verification (Testing)

- Similar to conventional software testing (SDLC).
- To ensure the developed KBS **meets the specification** and provides the proper solution to the problem.
- **Involves comparing the system** to its specification and testing the operation of the system.

KBS Life Cycle Model

Design Adjustment

- Some **adjustments made** to the design of the system.
- As long as the adjustments are relatively **minor** in nature and not retroactive, this is good.
- If, however, the **design changes are significant in scope and/or retroactive** to what has already been developed and validated, this can be a setback to the project, resulting in a paradigm shift.

KBS Life Cycle Model

Maintenance

- This stage is quite similar the conventional software life cycle (SDLC).
- This may involves in **adding/deleting knowledge representation**, upgrading the tool's or software's version, or editing system interface.

Knowledge Acquisition



To compile a body of knowledge.

Sources of knowledge: books, reports, database reports, The Internet.



Acquiring knowledge from the expert – knowledge elicitation.

Knowledge Acquisition Technique

- Eliciting knowledge **from an expert** can involve long and tedious session between KE and the expert.
- The session may be an interactive discussion that involve an exchange of ideas about the problem.
- Interviews are the most common, popular and traditional techniques in KA process.
- Others such as protocol analysis, repertory grid analysis and case study.



Knowledge Based System

Knowledge Acquisition Technique

- The **protocol analysis** involves in requiring the expert to **thinks-aloud** while working through a series of either simulated or written case examples.
- The repertory grid analysis produces **a set of dimensions defining the space** which contains the domain objects. Knowledge Engineer decides on a list of elements first and then presents them in trees asking the expert to indicate how one of them differs from the other – **Clustering information**.

Knowledge Acquisition Technique

- The case study tries to uncover the knowledge by **watching** the expert solve a real problem.
- Automatic induction tools are introduced to automate the knowledge acquisition process.



SUMMARY

KBS Life Cycle

- The waterfall model is used for developing an **ordinary software**.
- The **KBS lifecycle** is a methodology used for **developing KBS**.
- Some **differences** between the waterfall and KBS Lifecycle.
- **Knowledge acquisition** is an important phase in KBS lifecycle.